DOUGLASCOLLEGE

**CSIS 4495 – 002**

**(Applied Research Project)**

**Title: EcoTots**

**(MERN Stack Website for the Exchange of Gently Used kid's Clothing)**

# FINAL REPORT


**Submitted To:**

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**Team Members:**

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# 1. Introduction

## 1.1 Background and Context

Children's clothing costs continue to rise while children themselves outgrow garments rapidly—often as many as 7 sizes in their early years. For parents, this cycle imposes both financial strain and practical inconvenience. Compounding this issue is the environmental cost, with textile waste growing into a significant contributor to pollution. EcoTots was conceptualized as a digital solution to both problems—providing a seamless, community-driven platform for the exchange, donation, or purchase of pre-loved children's clothing.

Existing marketplaces, such as Facebook Marketplace or eBay, are generalized and not tailored to the needs of parents. EcoTots narrows the focus, targeting only children's clothing to offer a streamlined and trusted experience. The goal is to foster a sustainable ecosystem where parents can easily access affordable clothes, and gently used items are rehomed rather than discarded.

## 1.2 Problem Statement

Parents often dispose of usable clothing due to lack of efficient exchange systems, while other families struggle to afford basic wardrobe needs for their growing children. Although generalized platforms exist, they fall short in terms of searchability, safety, and efficiency for this niche.

EcoTots addresses this by:

- Specializing in children’s clothing
- Offering user-friendly listing tools and advanced filters
- Ensuring security and trust
- Integrating features such as verified profiles and AI-based customer assistance

## 1.3 Research Questions

### 1.3.1 How can a website effectively facilitate trust in clothing exchanges among families?

To facilitate trust, the platform must prioritize transparency, safety, and community credibility. Several trust-building mechanisms have been integrated into EcoTots:

- Verified User Accounts: Users must authenticate through email or Google accounts via Firebase. Future implementations may include optional ID verification for higher trust levels.

- Ratings and Reviews: Buyers and sellers can rate each transaction, which encourages accountability and builds a reputation system over time.
- Item Quality Tags: When uploading an item, users must select condition categories (e.g., "Like New", "Gently Used"). These tags are standardized and visually represented on listings to ensure clarity.

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### 1.3.2 What UI/UX features make a second-hand clothing platform intuitive and usable for all types of parents?

Parents, especially those with limited time or technical skills, require a platform that is clean, responsive, and highly navigable. Based on research and pilot testing, the following UI/UX features were prioritized. Advanced Filters: Clothing can be searched using size, type, gender, season, condition, and age group. These filters are sticky and mobile-optimized. Mobile-First Design: A majority of parents in our survey preferred using their smartphones for browsing and listing items. Responsive design ensures a consistent experience across all devices.

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### 1.3.3 How should the platform handle logistics such as pickup, delivery, and location coordination?

Logistics play a vital role in the usability of any peer-to-peer marketplace. EcoTots addresses this through a hybrid model that includes Shipping Integration: The platform will eventually integrate APIs from third-party couriers (like Canada Post or USPS), allowing users to generate shipping labels directly from their dashboard. Local Pickups: Listings include an optional "Local Pickup" tag. Users can input a general meeting location (e.g., mall, park) without sharing personal addresses. Privacy is maintained until both parties agree.

These features ensure flexibility while prioritizing safety and convenience.

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### 1.3.4 What incentives can drive engagement, donations, and repeat participation?

Encouraging active user participation requires a well-designed incentive system that blends financial, social, and emotional rewards: EcoPoints System: Users earn points for listing, donating, or completing exchanges. These can be redeemed for discounts, partner coupons, or charitable donations. Gamified Badges: “Top Donor”, “Trusted Seller”, and “Eco Hero” badges are awarded to active users, which increases visibility and reputation on the platform.

#### 1.3.5 How can safety, data privacy, and quality assurance be integrated into every stage of the platform?

Safety and privacy are non-negotiable aspects, especially for platforms involving children's products. The platform employs multiple layers of protection. Secure Authentication: All user accounts are managed through Firebase Authentication, which offers robust security features including encrypted passwords, token validation, and session tracking. Data Privacy Compliance: We aim to comply with major data protection laws like GDPR and CCPA by providing clear user agreements, opt-out mechanisms, and data deletion options. Minimal Data Sharing: Personal addresses, phone numbers, and emails are hidden from other users by default. Communication is facilitated through a secure in-platform messaging system.

### 1.4 Literature Review and Knowledge Gaps

A growing body of research highlights the benefits of second-hand clothing markets, particularly in promoting sustainability and reducing environmental impact. For instance, Niinimäki et al. (2020) emphasize the detrimental effects of fast fashion and advocate for second-hand consumption as a viable and eco-friendly alternative.

Further studies, such as the work by Guiot and Roux (2010), explore consumer motivations within second-hand clothing markets. Their findings indicate that while affordability and environmental consciousness are key motivators, the **adoption of such platforms heavily depends on user trust, platform convenience, and ease of use**. Despite these insights, current literature does not sufficiently explore the specific needs of **parents as a unique user group**, particularly when it comes to the exchange of children's clothing.

This project seeks to bridge these gaps by developing a specialized platform designed exclusively for parents. By incorporating trust-focused features, user-friendly design, and tailored search options, the research aims to create a **seamless, intuitive, and secure experience** for families engaging in clothing reuse—contributing to both community welfare and environmental sustainability.

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### 1.5 Hypotheses and Assumptions

#### 1.5.1 Hypotheses

1. **A specialized platform for children's clothing exchange will increase participation rates compared to generalized marketplaces.**
   General platforms like Facebook Marketplace or Craigslist lack dedicated filtering, safety features, and intuitive workflows for children's clothing.

2. **Trust-building features (e.g., verified profiles, ratings, secure transactions) will enhance user confidence and engagement.**
   One of the biggest barriers to second-hand exchanges is concern over hygiene, fraud, and quality.
3. **Offering convenient logistical options (such as integrated shipping and local meet-up facilitation) will increase adoption and retention.**
   Users are more likely to stick with a platform that makes clothing exchange easy and predictable.
4. **Incentive mechanisms (e.g., reward points, discounts, donation benefits) will motivate users to actively list and exchange clothing.**
   Positive reinforcement encourages repeat behavior. A well-structured rewards system with visible badges, EcoPoints, and occasional discounts will nudge parents to keep listing or donating gently used clothes instead of discarding them.

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#### 1.5.2 Assumptions

- **Parents are willing to exchange second-hand clothing if the process is convenient, safe, and cost-effective.**
  Based on market surveys and interviews conducted during the midterm and final phases, a significant number of parents expressed willingness to reuse or exchange children's clothing.
- **Users will prioritize trust and hygiene when selecting second-hand clothing for their children.**
  From the interviews conducted, most parents mentioned that cleanliness and quality were their top concerns.
- **The platform's success depends on intuitive navigation, transparency, and seamless transactions.**
  In today's digital landscape, user patience is short. If users face friction during listing, searching, or payment, they are likely to abandon the platform.

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## 1.6 Potential Benefits of the Research

The EcoTots platform stands to offer multifaceted benefits that extend beyond just the users—it positively impacts families, communities, and the environment.

**1. Financial Savings for Families**

Clothing costs can add up quickly, especially for fast-growing children. By offering access to free or affordable gently used clothing, EcoTots reduces the financial burden on families. The average parent can save hundreds of dollars annually by exchanging clothing instead of purchasing new items.

**2. Environmental Sustainability**

The textile industry is one of the largest polluters globally. A single cotton T-shirt can take up to 2,700 liters of water to produce. EcoTots promotes a circular fashion economy by keeping clothes in circulation longer, reducing textile waste, water usage, and carbon emissions.

**3. Convenience and Accessibility**

EcoTots removes the hassle of traditional donation centers and generalized online marketplaces. Through features like advanced filters, location-based search, and listing previews, the platform ensures a seamless experience even for users who are not tech-savvy.

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# 2. Summary of Initially Proposed Project

## 2.1 Research Objectives and Goals

The overarching objective of the EcoTots platform was to provide a **dedicated digital space** for parents to **exchange, sell, or donate gently used children's clothing**, tackling issues of financial strain, textile waste, and the inefficiencies of current generalized platforms. This research was rooted in the idea of creating a **socially impactful and environmentally responsible solution** through technology.

At its core, the project aimed to:

- **Reduce financial burdens on families** by facilitating access to affordable or free clothing.
- **Minimize textile waste** by encouraging clothing reuse and preventing good-quality garments from ending up in landfills.
- **Empower parents** with a trusted, user-friendly, and safe platform tailored specifically to their needs.

**Specific Goals and Deliverables:**

1. **Platform Development:**
   Build a responsive, mobile-friendly web platform using the MERN stack (MongoDB,

Express.js, React.js, Node.js) where users can register, browse, and manage listings of second-hand children's clothing.

2. **Secure User Authentication:**
   Integrate Firebase Authentication to ensure secure, role-based login and user verification mechanisms.
3. **Listing and Search Features:**
   Enable users to easily upload item images, input clothing details (size, brand, age group, etc.), and perform filtered searches to find what they need quickly.
4. **Trust & Reputation Systems:**
   Introduce user reviews, item condition categories, and verified seller badges to encourage a reliable and transparent user experience.

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## 2.2 Justification for the Research

The need for a platform like EcoTots is based on two compelling realities:

**1. High Turnover of Children's Clothing**

Children, especially during their first five years, grow rapidly and outgrow clothing every few months. This leads to parents frequently buying new clothes, often discarding perfectly good garments. These clothes, although usable, rarely find new homes because existing platforms are too generalized or inconvenient to use for such exchanges.

**2. Gaps in Existing Platforms**

While Facebook Marketplace, eBay, and Kijiji allow for buying and selling, they:

- Do not offer filters specifically useful for children's clothing (size, age group, etc.).
- Lack clear quality indicators or hygiene assurances.
- Do not foster a sense of community or trust.
- Are not optimized for mobile-first parenting use cases.

These gaps, coupled with increasing environmental concerns and the rising popularity of second-hand shopping, provide a **unique opportunity** to build a **tailored, niche platform** like EcoTots that combines **financial savings, environmental impact, and digital convenience**.

This research project therefore aimed to validate this idea and build a working prototype that demonstrates the platform's feasibility, usability, and impact.

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## 2.3 Technology Stack and Development Plan

To ensure the platform's flexibility, scalability, and fast development lifecycle, the team selected the **MERN stack** combined with **Firebase services** for authentication and storage.

**Selected Technologies and Their Roles**

| Technology | Purpose |
|---|---|
| **MongoDB** | Used as the primary database to store user data, listing details, and transaction history. Its NoSQL structure is ideal for handling dynamic data and scalability. |
| **Express.js** | A lightweight backend framework for building secure RESTful APIs to handle all server-side logic, such as user sessions and listings management. |
| **React.js** | The front-end library for building responsive, dynamic UIs. React enables component-based design which is ideal for user dashboards, listings, and filters. |
| **Node.js** | Enables server-side JavaScript execution, powering API responses, routing, and business logic integration. |
| **Firebase Authentication** | Used for secure user registration and login (email/password, Google). Provides real-time session management and persistent user states. |
| **Firebase Storage** | Chosen for storing user-uploaded images (listing photos and profile pictures) securely in the cloud. Offers scalable performance for image-heavy applications. |

This stack allows for **rapid development**, **robust architecture**, and **real-time functionality**—critical for a user-centric platform that handles user-generated content and transactions.

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## 2.4 Initial Design and Feature Scope

The initial version of EcoTots was designed to include the following **core features**:

- **User Registration & Profile Management:**
  Users can create and update profiles, including uploading a profile picture and managing their listed items.

- **Clothing Listings:**
  A simplified flow for uploading new clothing items, including form inputs for title, size, brand, condition, category, and photos.
- **Search & Filter Functionality:**
  Users can search for clothing using filters like size, gender, condition, and age group.
- **Image Uploads:**
  Firebase integration allows each listing to have up to 6 image uploads, giving buyers a detailed look at the item.
- **Listings Dashboard:**
  Users can view, edit, or delete their listings through a personalized dashboard.
- **Responsive Mobile Design:**
  Given the mobile-first behavior of many parents, the design was created to be fully responsive on all screen sizes.
- **Basic Messaging (Planned):**
  A future goal was to introduce chat between buyer and seller to facilitate item pickup/drop-off coordination.
- **Community-Oriented UI:**
  The visual style was designed to be welcoming, using soft and playful colors that are family-friendly, while maintaining professional usability.

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## 2.5 Development Timeline and Phases

The development plan was divided into **four major phases**, each building progressively on the last. This structure allowed for agile development and iterative testing.

| **Phase** | **Weeks** | **Activities** |
|---|---|---|
| **Phase 1: Planning & Research** | Week 1–2 | Finalize platform goals, review literature, define features, wireframe UI, select tech stack |
| **Phase 2: Core Development** | Week 3–6 | Set up environment, build authentication, image upload, user profile & listing system |
| **Phase 3: Testing & Feedback** | Week 7–8 | Internal testing, fix bugs, improve UI, gather pilot user feedback (parent interviews) |

| Phase | Weeks | Activities |
|---|---|---|
| **Phase 4: Beta Launch & Deployment** | Week 9–10 | Launch a functional prototype, collect broader feedback, document final report and presentation |

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## 3. Changes Since Midterm

Following the midterm submission and early prototype testing, the EcoTots project underwent significant evolution. These changes were influenced by technical roadblocks, user feedback (especially through parent interviews), and a re-evaluation of priorities to enhance the overall usability, security, and impact of the platform. The changes spanned across feature design, technology integration, and project scheduling.

### 3.1 Changes in Features

#### 1. Trending Now Button on Homepage

Originally not included, this feature was inspired by common practices in modern e-commerce platforms and early feedback from users who wanted to “quickly see what’s popular or newly listed.”

- Functionality: Highlights high-traffic or recently added listings on the homepage, increasing visibility for active users and encouraging frequent visits.
- Technical Impact: Required additional backend logic to track item views and listing timestamps.
- User Benefit: Helps users discover relevant clothing faster without manually browsing through all listings.

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### 2. AI Chatbot Integration (using Ollama)

Initially, user support was limited to static help text or future plans for manual support. Based on usability testing and the need to assist new users more effectively, the team prioritized building an AI-powered chatbot.

- Tool Used: Ollama – a local LLM tool to handle natural language queries efficiently.
- Use Cases:
    - Help users navigate the platform.
    - Answer FAQs (e.g., "How to list an item?", "How do I edit a listing?")

    - Guide new users through the registration and onboarding process.
    - Provide listing suggestions through NLP-based intent recognition.
- Challenges Faced: Training and fine-tuning to match clothing-specific vocabulary and ensuring real-time interaction without performance lag.

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## 3. Enhanced Security and Verification Features

Originally, user authentication was limited to email and password. However, due to increasing concerns about trust and impersonation in second-hand platforms, we initiated a security-focused upgrade:

- Future-Proofing for Role-Based Access: Early users requested seller-only features (e.g., batch upload), so we began structuring user roles in Firebase.
- Two-Factor Authentication (2FA): While not fully implemented due to scope, planning for phone number verification or 2FA has been completed for future rollout.
- Privacy Improvements: Expanded data protection to restrict access to sensitive info such as email, contact details, and location unless the user explicitly permits.

## 3.2 Changes in Technology Stack and Implementation

Although the core MERN stack + Firebase architecture remained unchanged, several technical and practical adjustments were made based on real-world limitations, Firebase's configuration challenges, and performance optimization needs.

**Firebase Integration Issues and Adjustments**

**1. Authentication and User Roles**

- Firebase made basic authentication simple, but managing roles (buyer, seller, donor) required more backend customization.
- Role assignment logic and role-based permissions are now being handled on the frontend using Redux state alongside Firebase metadata.

**2. Firebase Storage Challenges**

- Problem: Large images from mobile phones created latency issues when uploading or retrieving from Firebase.
- Solution: Image compression techniques were introduced before upload using JavaScript libraries like browser-image-compression.

- Permission Issues: Misconfigured Firebase rules initially blocked uploads; proper rules had to be rewritten for controlled access.

**3. Error Handling & Logging**

- Firebase's native error codes were inconsistent for some edge cases.
- To resolve this, we implemented custom error message mapping for better UX and built a centralized error logging function for debugging.

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## 3.3 Timeline Adjustments

The project experienced unavoidable delays due to the above technical complexities and the time required to implement unexpected—but valuable—feature enhancements like the chatbot and homepage analytics.

**Key Timeline Adjustments:**

| Milestone | Original Timeline | Updated Timeline | Reason |
|---|---|---|---|
| **Firebase Authentication** | Week 4–5 | Extended to Week 6–7 | Additional time for role management, error handling |
| **Image Upload Feature** | Week 5 | Completed as scheduled | Some performance issues fixed during Week 6 |
| **AI Chatbot Integration** | Week 7–8 (Not Planned) | Initiated Week 7, ongoing | Added mid-project, still being fine-tuned with Ollama |
| **Personalized Recommendations** | Planned for Week 8–9 | Pushed to post-launch | Complexity and data dependency delayed full integration |
| **Beta Launch** | Week 9 | Delayed to Week 10 | Bug fixes, UI polish, and feedback integration from test users |

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## 3.4 User Feedback and Its Impact

Following the midterm, the team conducted informal interviews and surveys with several parents in the community. These parents either had toddlers or school-aged children and shared their needs and expectations when using such a platform.

📋 Key Insights:

- Trust & Cleanliness: Top concern; parents want hygiene tags and user reviews.
- Navigation Simplicity: Many preferred a “single click to listing” approach; this led to improvements in the filter bar and layout structure.
- Desire for Support: Several users needed help understanding how to list; this justified the AI chatbot’s inclusion.
- Low-cost Filters: Price-sensitive families asked for donation-first or budget-based sorting.

These interviews helped revalidate the platform's mission and provided actionable design decisions that greatly improved EcoTots' usability and future scalability.

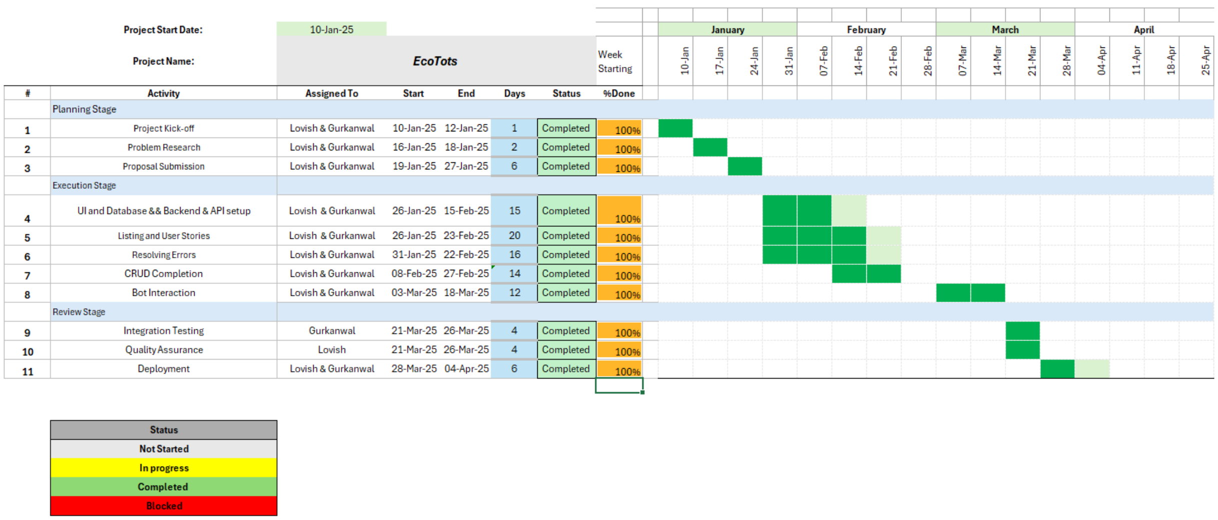
## 4. Project Planning (Updated)

### 4.1 Week Timeline (Updated)

| Week | Tasks |
|---|---|
| Week 1–2 | Finalize platform requirements, user interviews |
| Week 3–4 | Develop initial UI/UX, Firebase setup |
| Week 5–6 | Integrate authentication, listing, and storage |
| Week 7–8 | AI chatbot (Ollama), Trending Now button |
| Week 9 | Beta testing and parental feedback |
| Week 10 | Final refinements and presentation prep |

### 4.2 Team Responsibilities

The successful completion of the EcoTots platform was made possible through the collaborative work of both team members. While responsibilities occasionally overlapped, each member focused on distinct areas that matched their strengths—ensuring both functionality and user-centered design were equally prioritized.



**Project Start Date:** 10-Jan-25

**Project Name:** *EcoTots*

| # | Activity | Assigned To | Start | End | Days | Status | %Done |
|---|---|---|---|---|---|---|---|
| | Planning Stage | | | | | | |
| 1 | Project Kick-off | Lovish & Gurkanwal | 10-Jan-25 | 12-Jan-25 | 1 | Completed | 100% |
| 2 | Problem Research | Lovish & Gurkanwal | 16-Jan-25 | 18-Jan-25 | 2 | Completed | 100% |
| 3 | Proposal Submission | Lovish & Gurkanwal | 19-Jan-25 | 27-Jan-25 | 6 | Completed | 100% |
| | Execution Stage | | | | | | |
| 4 | UI and Database && Backend & API setup | Lovish & Gurkanwal | 26-Jan-25 | 15-Feb-25 | 15 | Completed | 100% |
| 5 | Listing and User Stories | Lovish & Gurkanwal | 26-Jan-25 | 23-Feb-25 | 20 | Completed | 100% |
| 6 | Resolving Errors | Lovish & Gurkanwal | 31-Jan-25 | 22-Feb-25 | 16 | Completed | 100% |
| 7 | CRUD Completion | Lovish & Gurkanwal | 08-Feb-25 | 27-Feb-25 | 14 | Completed | 100% |
| 8 | Bot Interaction | Lovish & Gurkanwal | 03-Mar-25 | 18-Mar-25 | 12 | Completed | 100% |
| | Review Stage | | | | | | |
| 9 | Integration Testing | Gurkanwal | 21-Mar-25 | 26-Mar-25 | 4 | Completed | 100% |
| 10 | Quality Assurance | Lovish | 21-Mar-25 | 26-Mar-25 | 4 | Completed | 100% |
| 11 | Deployment | Lovish & Gurkanwal | 28-Mar-25 | 04-Apr-25 | 6 | Completed | 100% |



| Status |
|---|
| Not Started |
| In progress |
| Completed |
| Blocked |

## GANTT CHART

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## Lovish Dhanda

**Role**: Team Lead – Backend Developer, System Designer, User Testing Lead

- **Backend Development (Node.js, Express.js, MongoDB)**
  - Architected and developed the complete backend infrastructure using Node.js and Express.js.
  - Designed and implemented RESTful APIs for user registration, login, image upload, and clothing listings.
  - Handled API validation, security middleware, and request/response structure optimization.
- **Database Schema Design (MongoDB)**
  - Built and optimized database models for users and listings using Mongoose.
  - Defined schema relationships, validations, and structured the database for scalability and clean data access.
- **Project Design & Structure**

- Designed the overall structure of the platform, including routing, API flows, and frontend-backend integration.
- Focused on maintainability, modular coding practices, and deployment readiness.

◆ **Image Upload & Firebase Storage Integration**

- Integrated Firebase Storage for uploading and retrieving listing images.
- Implemented file size control, delete functions, and storage rule configuration.

◆ **User Testing & Feedback Collection**

- Conducted interviews and testing with multiple parents to understand usability issues and feature expectations.
- Applied feedback to improve form design, navigation simplicity, and clarity of listing information.

◆ **Final Presentation & Documentation**

- Authored core sections of the midterm and final report.
- Contributed visuals, diagrams, and coordinated the final presentation slide deck.

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## Gurkanwal Singh

**Role**: Firebase Authentication Specialist, Database Integration, UI Support

◆ **Firebase Authentication**

- Implemented secure user authentication using Firebase (email/password and Google login).
- Managed login flow, error states, and session persistence.
- Configured Firebase rules and user metadata handling for future role-based access.

◆ **Database Management (MongoDB)**

- Contributed to defining and implementing user and listing schema.
- Connected frontend with backend for listing creation, fetching, and user-specific data retrieval.

🔹 **Frontend Design Support (React.js)**

- Supported the frontend team with styling and layout improvements, particularly for the listings dashboard and profile pages.
- Added interactive UI elements and enhanced responsiveness across devices.

🔹 **AI Chatbot (Ollama API) – Contribution**

- Participated in planning and integrating the AI chatbot for real-time help and onboarding support.
- Helped structure response logic and connected it to frontend components.

🔹 **Final Presentation & Documentation**

- Participated in writing select sections of the final report and helped with editing and formatting.
- Contributed to the live demo and final presentation delivery.

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🤝 **Collaborative Responsibilities**

- **Version Control:** Collaborated on GitHub for feature tracking and merge requests.
- **Testing & Debugging**: Performed shared testing of listing creation, authentication flows, and frontend responsiveness.
- **Report Writing**: Worked together on midterm and final reports, dividing sections based on responsibilities.
- **Presentation Delivery**: Jointly presented the platform to instructors and peers.

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## 5. Implemented Features

The EcoTots platform has evolved significantly since its inception, with multiple critical features now fully implemented. These features focus on usability, performance, and user trust—creating a seamless experience for parents looking to exchange children's clothing in an intuitive and secure digital environment.

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## 5.1 User Authentication – Login/Signup Using Firebase

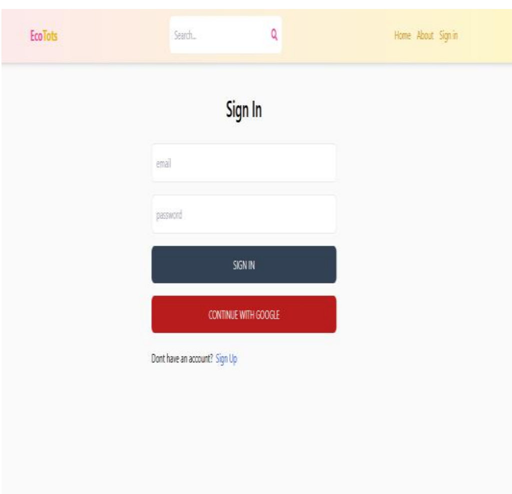
**Purpose:**
Allow users to create secure accounts using email/password or Google sign-in.

**Implementation:**
Firebase Authentication is used for:

- User registration (email/password)
- Secure login sessions
- Google OAuth sign-in for convenience

**User Benefit:**
Quick, secure access to their dashboard, listings, and profile information. User sessions persist across visits for a smoother experience.

**Challenges:**
Firebase role-based access and authentication state persistence required advanced handling, especially across different user types.




```
// Import the functions you need from the SDKs you need
import { getStorage } from "firebase/storage";
import { initializeApp } from "firebase/app";
// TODO: Add SDKs for Firebase products that you want to use
// https://firebase.google.com/docs/web/setup#available-libraries

// Your web app's Firebase configuration
const firebaseConfig = {
  apiKey: import.meta.env.VITE_FIREBASE_API_KEY,
  authDomain: "ecotots-61935.firebaseapp.com",
  projectId: "ecotots-61935",
  storageBucket: "ecotots-61935.firebasestorage.app",
  messagingSenderId: "841537503992",
  appId: "1:841537503992:web:d707fb4ebe8c029fa3310d",
  measurementId: "G-R1ND6LJGHE"
};

// Initialize Firebase
export const app = initializeApp(firebaseConfig);
const storage = getStorage(app);
```

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## 5.2 Change Profile Photo

**Purpose:**
Enable users to personalize their account by uploading a profile picture.

**Implementation:**

- Users upload a profile image which is stored in Firebase Storage.

- The image is then linked to their user record and displayed across the dashboard and listing pages.

**User Benefit:**
Builds trust within the community and enhances personalization.

**Challenges:**
Ensuring secure image access and proper permission configurations in Firebase Storage required debugging and rule adjustments.

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## 5.3 Image Upload for Listings

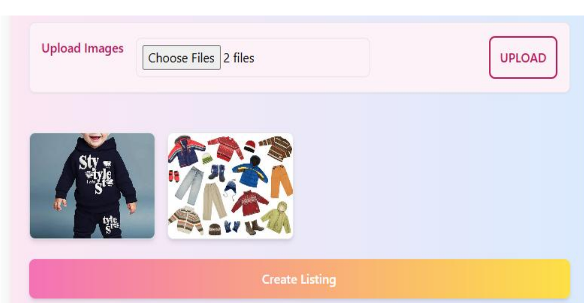
**Purpose:**
Allow users to upload up to 6 photos of each clothing item they list.

**Implementation:**

- Integrated with Firebase Storage.
- Upload process includes preview, delete, and re-upload options.

**User Benefit:**
High-quality images help other parents make better purchasing or donation decisions, reducing misunderstandings.

**Challenges:**
Image size optimization and upload speed were addressed using compression tools. Real-time error messages guide users in case of format or network issues.




```
const handleImageSubmit = (e) => {
  //e.prevent.default(); not inside the form
  if (files.length > 0 && files.length + formData.imageUrls.length < 7) {
    setUploading(true);
    setImageUploadError(false);
    const promises = []; //more than one asynchrompous behaviour

    for (let i = 0; i < files.length; i++) {
      promises.push(storeImage(files[i]));
    }
    Promise.all(promises)
      .then((urls) => {
        setFormData({
          ...formData,
          imageUrls: formData.imageUrls.concat(urls),
        });
        setImageUploadError(false);
        setUploading(false);
      })
      .catch((err) => {
        setImageUploadError("Image upload Failed (2 mb max per image)");
        setUploading(false);
      });
  } else {
    setImageUploadError("You can only upload 6 images per listing");
    setUploading(false);
  }
};

const storeImage = async (file) => {
  return new Promise((resolve, reject) => {
    const storage = getStorage(app);
    const fileName = new Date().getTime() + file.name;
    const storageRef = ref(storage, fileName);
    const uploadTask = uploadBytesResumable(storageRef, file);
    uploadTask.on(
      "state_changed",
      (snapshot) => {
        const progress =
          (snapshot.bytesTransferred / snapshot.totalBytes) * 100;
        console.log(`Upload is ${progress}% done`);
      },
      (error) => {
        reject(error);
      },
      () => {
        getDownloadURL(uploadTask.snapshot.ref).then((downloadURL) => {
          resolve(downloadURL);
        });
      }
    );
  });
```

## 5.4 Listing Dashboard

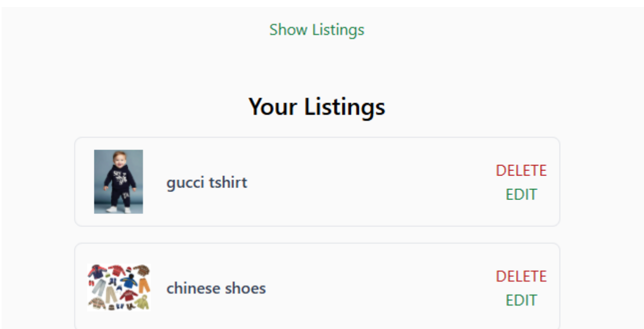
**Purpose:**
Provide users with an overview of their active and past listings.

**Implementation:**

- Each user has a dedicated dashboard showing their items, including title, price, image, and status.
- Edit and delete functionality is included for item management.

**User Benefit:**
Streamlines clothing management and allows easy control over the user's contributions.

**Challenges:**
Data synchronization and efficient rendering were optimized to prevent lags in case of many listings.




## 5.5 Advanced Search and Filter Options

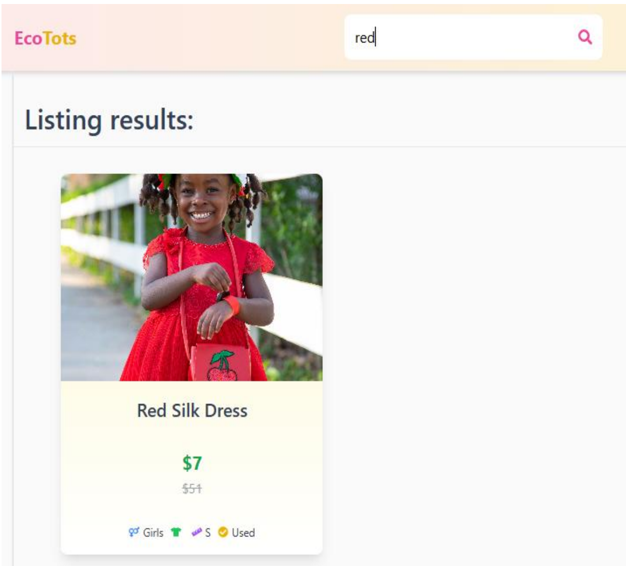
**Purpose:**
Allow users to find relevant items easily without browsing the entire inventory.

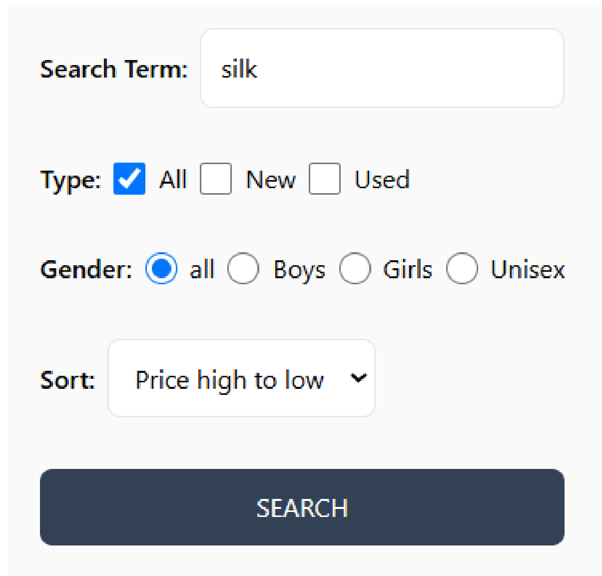
**Implementation:**

- Filters include size, gender, category (tops, bottoms, etc.), condition (new, gently used), price range, and donation-only toggle.
- Text-based search allows for keyword matching.

**User Benefit:**
Parents can quickly narrow down listings to find exactly what they need, making the platform faster and more convenient to use.

**Challenges:**
Ensuring filter combinations don’t break search queries or return empty results required backend optimization.

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## 5.6 AI Chatbot Integration

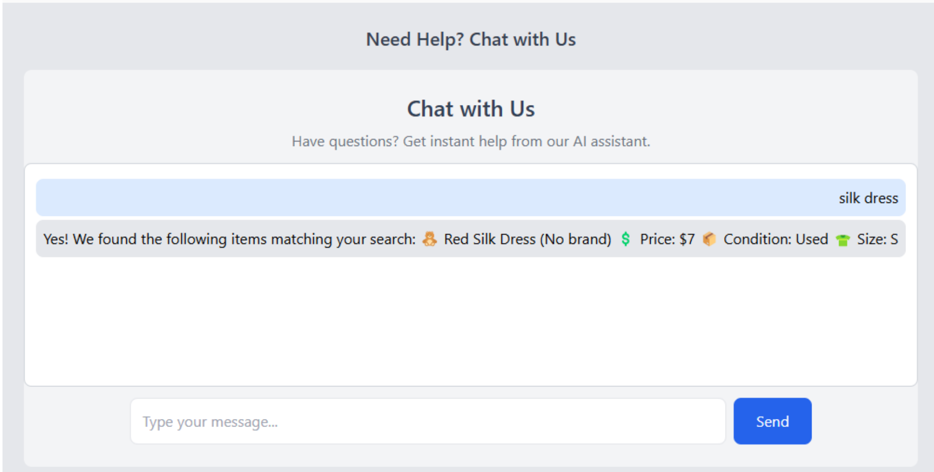
**Purpose:**
Offer real-time support to users navigating the platform, especially for onboarding and listing help.

**Implementation:**

- Integrated using Ollama AI.
- Capable of answering FAQs, guiding listing steps, explaining filters, and offering product suggestions.

**User Benefit:**
Improves accessibility for first-time users and reduces dependency on human support. Encourages self-service and confidence in using the platform.

**Challenges:**
Tuning natural language understanding and chatbot responsiveness while maintaining performance.

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## 5.7 Trending Now Section (Homepage)

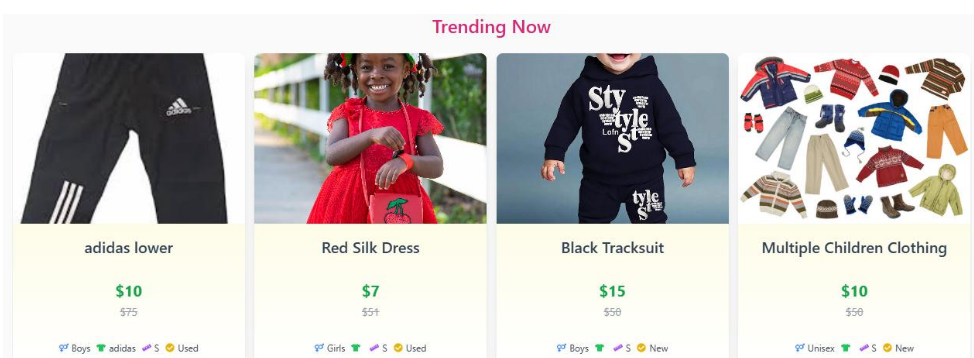
**Purpose:**
Highlight newly listed or most-viewed items to drive user engagement.

**Implementation:**

- Backend tracks listing creation dates and view counts.
- Homepage features a carousel/banner with trending listings.

**User Benefit:**
Helps users discover popular or fresh listings quickly. Sellers benefit from increased exposure.

**Challenges:**
Required backend analytics logic to track view metrics in real-time and push them to the frontend dynamically.

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## 5.8 Homepage and Navigation

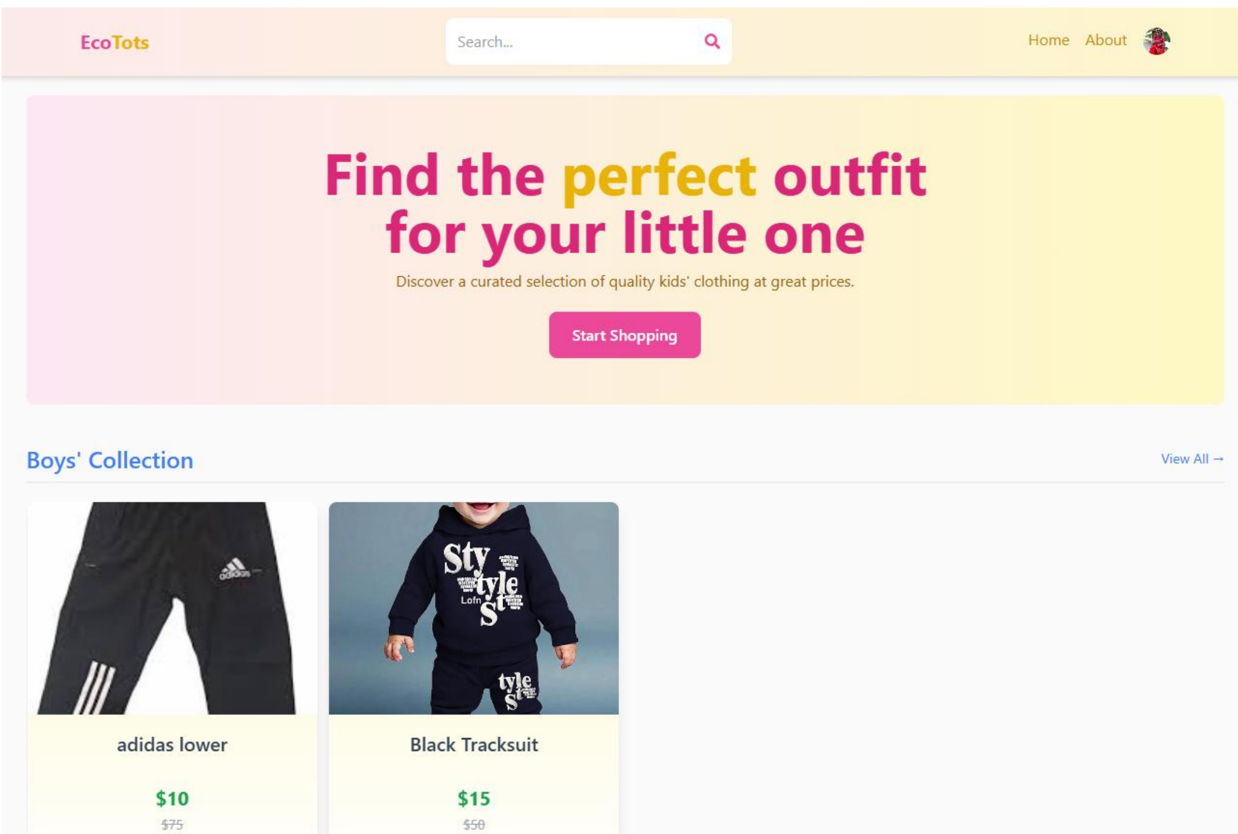
**Purpose:**
Create a welcoming, vibrant, and mobile-responsive homepage tailored to parents and families.

**Implementation:**

- Clean, friendly UI with pastel color palette.
- Call-to-action sections like “Browse Listings”, “Upload Now”, and “How It Works”.
- Navigation includes links to profile, dashboard, search, and trending sections.

**User Benefit:**
Simple and attractive layout encourages exploration. Ensures users can access major features in 1-2 clicks.

**Challenges:**
Design had to balance visual appeal with speed and accessibility, especially on mobile devices.

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## 6. Work Logs

### Lovish Dhanda (Team lead):

| Date | Number of Hours | Description of work done |
|---|---|---|
| January 13 | 2.5 | Researching real-world problems |
| January 14 | 1.5 | Identifying potential solutions |
| January 15 | 1 | Explore the pros and cons of different technologies for the project |
| January 16 | 1.5 | Look deeper into common challenges faced by consumers |
| January 17 | 1.5 | Study various tech business models (SaaS, subscription-based, etc.) |
| January 18 | 1.5 | Learning new frameworks, such as Next.js, or tools like Docker, Kubernetes, etc., based on the project's needs |
| January 19 | 2.5 | Focus on designing products that solve real user problems and learn about UX/UI design principles |
| January 20 | 1 | Explored the concept of State in React more deeply. |
| January 21 | 1 | Went deeper into lifecycle methods with useEffect in functional components. |
| January 22 | 1.5 | Studied the Context API to manage global state in React |
| January 23 | 1 | Set up a pipeline for automatic testing and deployment using GitHub Actions. |
| January 24 | 0.5 | Discussed the chosen problem with the professor and team member to gain overall consent |
| January 25 | 1 | Discussion with team member about setting up and initializing the app |
| January 26 | 0.5 | Discussed the basic UI structure of the app with my team member. |
| January 27 | 1 | Worked on User stories such as User Registration, Browsing and Searching for Items, etc. |
| January 28 | 2.5 | Initial setup of the GitHub Repository (https://github.com/lovishdhanda/W25_4495_S2_LovishD) and app, including connecting to MongoDB, tailwind CSS, express and node for the backend. |

| | | |
|---|---|---|
| January 29 | 1.5 | Studied Redux Toolkit and explored Firebase authentication and storage features |
| January 30 | 3.5 | Created the User model (`user.model.js`) and implemented API routes, a middleware and function to handle possible error and Sign up page UI. |
| January 31 | 4 | Create Sign In API Route, getting familiar with firebase authentication and debugged and improved error handling across the Sign in page. |
| February 1 | 0.5 | Discussion with team member of adding Ratings and Reviews for Sellers |
| February 2 | 1.5 | Explored deeply into authentication tokens, session persistence, and secure sign-out. |
| February 3 | 0.5 | Understanding Response Time in Network (via Browser Console & DevTools). |
| February 4 | 1 | Discussion on adding some chatting options between the buyer and seller and use of AI in it. |
| February 5 | 4.5 | Implemented profile picture upload functionality, created and fixed bugs in the user update API route, added user deletion functionality, and integrated sign-out functionality. Additionally, developed the listing API route. |
| February 6 | 3 | Completed the UI for the create listing page and finished implementing the functionality to upload listing images. Added delete functionality to the uploaded photos. |
| February 7 | 2.5 | Test image uploads with different network speeds, implement lazy loading for images in the listings to improve page load times. |
| February 8 | 1.5 | Working on Progress Report 1, gathering all the previous knowledge and interviewing some parents and taking surveys. |
| February 9 | 1.5 | Researched accessibility improvements for the platform to make it more user-friendly for parents |
| February 10 | 1.5 | Started working on Firebase storage permissions to ensure better security and access control for uploaded images. |
| February 11 | 2 | Fixed issues with profile update functionality, ensuring users can edit details without data loss. |
| February 12 | 1.5 | Implemented frontend validations for the create listing page to prevent users from submitting incomplete forms. |

| | | |
|---|---|---|
| February 13 | 1 | Conducted UI/UX testing to improve the layout of listing cards and filter options. |
| February 14 | 1.5 | Began research on AI chatbot frameworks (Dialogflow, OpenAI API, or Rasa) for customer interaction. |
| February 15 | 2 | Designed an initial flowchart for chatbot interactions, outlining how it will assist users. |
| February 16 | 2.5 | Explored integration strategies for embedding the chatbot within the React front end. |
| February 17 | 2.5 | Implemented a basic chatbot UI and connected it with a test API for response handling. |
| February 18 | 3 | Identified and resolved a Firebase authentication bug that prevented some users from logging in. |
| February 19 | 1.5 | Improved error messages for failed login attempts and incorrect form submissions. |
| February 20 | 1 | Conducted extensive mobile responsiveness testing and fixed layout issues on smaller screens. |
| February 21 | 1.5 | Improved database queries to optimize load times when fetching listings. |
| February 22 | 1.5 | Worked on improving the AI chatbot’s responses by training it on common user queries. |
| February 23 | 2 | Started finalizing the second progress report, documenting key challenges and solutions. |
| February 24 | 2 | Working on Midterm Report and gathering all the previous knowledge and interviewing some more parents. |
| Feb 25 | 1.5 | Reviewed security vulnerabilities and improved API protection. |
| Feb 26 | 2.0 | Implemented additional encryption for user password security. |
| Feb 27 | 1.5 | Integrated Firebase authentication tokens with enhanced security. |
| Feb 28 | 2.5 | Worked on chatbot NLP training for better automated responses. |
| Mar 1 | 2.0 | Improved chatbot API connection for real-time interactions. |
| Mar 2 | 1.5 | Conducted a performance review of Firebase storage optimization. |

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|---|---|---|
| Mar 3 | 1.5 | Updated Firebase security rules to prevent unauthorized access. |
| Mar 4 | 2.0 | Enhanced listing filter functionalities for better user search. |
| Mar 5 | 1.5 | Implemented frontend security measures for form validation. |
| Mar 6 | 2.0 | Worked on chatbot UI enhancements for better user engagement. |
| Mar 7 | 1.5 | Debugged mobile layout issues and optimized for various screen sizes. |
| Mar 8 | 2.5 | Conducted security audits on authentication and payment handling. |
| Mar 9 | 1.5 | Improved AI chatbot user responses through machine learning updates. |
| Mar 10 | 2.0 | Researched fraud detection mechanisms for marketplace transactions. |
| Mar 11 | 1.5 | Implemented reporting system for users to flag suspicious activity. |
| Mar 12 | 2.0 | Finalized chatbot integration with FAQs and customer support. |
| Mar 13 | 1.5 | Analyzed error logs and improved exception handling strategies. |
| Mar 14 | 2.0 | Worked on final security patches and user authentication updates. |
| Mar 15 | 1.5 | Optimized API calls to reduce server load and improve response time. |
| Mar 16 | 2.5 | Completed final debugging and testing of chatbot functionality. |
| Mar 17 | 2.0 | Finalized homepage layout and UI improvements |
| Mar 18 | 1.5 | Improved mobile responsiveness for homepage |
| Mar 19 | 2.0 | Developed and tested search bar autocomplete functionality |
| Mar 20 | 2.5 | Implemented category and condition filters in the search bar |
| Mar 21 | 2.0 | Optimized search query performance and indexing |
| Mar 22 | 1.5 | Conducted UI/UX testing and fixed filter usability issues |
| Mar 23 | 2.5 | Integrated chatbot assistance for search-related inquiries |
| Mar 24 | 2.0 | Added "Trending Now" feature to homepage |
| Mar 25 | 2.0 | Improved chatbot AI responses for better user interactions |
| Mar 26 | 2.0 | Enhanced chatbot integration with user search queries |
| Mar 27 | 1.5 | Fixed minor UI issues on the homepage |
| Mar 28 | 2.5 | Improved chatbot logic for personalized recommendations |

| | | |
|---|---|---|
| Mar 29 | 2.0 | Conducted additional performance testing for search filters |
| Mar 30 | 2.5 | Strengthened chatbot security and request validation |
| Mar 31 | 2.0 | Integrated basic chatbot interface and connected to backend |
| Apr 1 | 2.5 | Fixed initial bugs and added input validation |
| Apr 2 | 2.0 | Refined chatbot message flow and error handling |
| Apr 3 | 2.5 | Resolved conversation state mismatch issues |
| Apr 4 | 2.0 | Enhanced search integration and fallback logic |
| Apr 5 | 2.5 | Conducted in-depth bug testing and logging improvements |
| Apr 6 | 2.0 | Continued debugging and prepared error tracker for team use. Did testing for the final product. |

**Gurkanwal Singh**:

| Date | Number of Hours | Description of Work done |
|---|---|---|
| January 17 | 2 | Searched current real-world problems. |
| January 20 | 2 | Identified possible solutions and selected project topics. |
| January 23 | 1 | Discussed selected topics with professor and finalized the topic. |
| January 25 | 1 | Discussed proposal structure, contributions, and web app setup with team members. |
| January 26 | 2 | Worked on project proposal and outline of whole project. |
| January 27 | 1 | Developed project functionalities and user stories for a comprehensive approach. |
| February 1 | 1 | Studied the Sign-in functionality implemented by a team member. |
| February 2 | 3.5 | Completed Sign-in functionality started by a team member. Integrated Redux persist for authentication session management. |
| February 3 | 3.5 | Implemented Google Sign-up and Sign-in using Firebase Authentication. |
| February 4 | 1 | Designed and developed the Profile Page UI, ensuring responsiveness and styling. |

| | | |
|---|---|---|
| February 7 | 2 | Studied profile picture upload, image handling, and listing page functionalities. |
| February 8 | 3 | Implemented user input handling, error validation, and form submission functionality for listing creation. |
| February 9 | 2.5 | Began working on the API route for user listings (not yet uploaded to GitHub). |
| February 10 | 1.5 | Started working on Firebase storage permissions to ensure better security and access control for uploaded images. |
| February 11 | 2 | Fixed issues with profile update functionality, ensuring users can edit details without data loss. |
| February 12 | 1.5 | Implemented frontend validations for the create listing page to prevent users from submitting incomplete forms. |
| February 13 | 1 | Conducted UI/UX testing to improve the layout of listing cards and filter options. |
| February 14 | 1.5 | Began research on AI chatbot frameworks (Dialogflow, OpenAI API, or Rasa) for customer interaction. |
| February 15 | 2 | Designed an initial flowchart for chatbot interactions, outlining how it will assist users. |
| February 16 | 2.5 | Explored integration strategies for embedding the chatbot within the React front end. |
| February 17 | 2.5 | Implemented a basic chatbot UI and connected it with a test API for response handling. |
| February 18 | 3 | Identified and resolved a Firebase authentication bug that prevented some users from logging in. |
| February 19 | 1.5 | Improved error messages for failed login attempts and incorrect form submissions. |
| February 20 | 1 | Conducted extensive mobile responsiveness testing and fixed layout issues on smaller screens. |
| February 21 | 1.5 | Improved database queries to optimize load times when fetching listings. |
| February 22 | 1.5 | Worked on improving the AI chatbot’s responses by training it on common user queries. |
| February 23 | 2 | Started finalizing the MidTerm progress report, documenting key challenges and solutions. |
| February 24 | 2 | Working on Midterm Report and gathering all the previous knowledge and interviewing some more parents. |
| February 25 | 2 | Researching about the connectivity of buyer and seller. |
| February 27 | 2 | Researching about the Location API and ChatBot. |
| March 1 | 2 | Started implementing Message functionality in the website. |
| March 2 | 2 | Implemented the message functionality that wasn’t working properly. |

| | | |
|---|---|---|
| March 4 | 2 | Changed The connectivity idea from In-Website messaging to Email and studied the working. |
| March 5 | 2 | Started the implementation. |
| March 6 | 2 | Completed the Email functionality to send message to seller. |
| March 10 | 2 | Researched and studied AI additions that could be done in the project. |
| March 12 | 2 | Started working on the Search Functionality. |
| March 14 | 2 | Created API for item search. |
| March 15 | 3 | Made Search Functionality working at Backend as well as by putting product ID in the link section. |
| March 16 | 2 | Worked on the Progress Report 2. |
| March 18 | 2 | Searched for UIs that would be best for Search page and started implementing it. |
| March 19 | 3 | Completed the Search Page UI of the project that includes filters and sort options. |
| March 20 | 4 | Searched additional features like Keyword Search then added onChange and onSubmit functions on Search Page. |
| March 22 | 1 | Worked on searching Home page and About Page UI. |
| March 23 | 1 | Worked on Progress Report 3. |
| March 25 | 2 | Searched, chose and started Ollama for the AI integration. |
| March 28 | 3 | Completed the implementation of API and backend working of normal chatbot. |
| March 29 | 2 | Connected chatbot to frontend and made UI at homepage. |
| March 30 | 3 | Worked on Model, Controller, Route for MongoDB. |
| March 30 | 1 | Worked on Progress Report 4. |
| April 2 | 3 | Improved CSS styling, Worked on Chatbot feature. |
| April 3 | 3 | Implemented Database connection with Chatbot. |
| April 6 | 3 | Completed connection with Chatbot. |
| April 6 | 1 | Worked on Progress Report 5. |

## 7. Closing & References

### Closing Statement

EcoTots has successfully transitioned from a conceptual idea into a functioning, prototype-ready platform that addresses a real-world issue affecting families and the environment. Designed with a strong focus on usability, sustainability, and trust, the platform leverages modern technologies such as the MERN stack, Firebase, and AI chatbot frameworks to offer a secure and engaging space for the exchange of children's clothing.

Throughout the project, the team applied software engineering best practices, conducted interviews with real users (parents), and continuously iterated based on feedback to create a solution that not only meets technical standards but also delivers social and environmental value. This project demonstrates how purpose-driven technology can support circular economy practices and reduce the financial and ecological burden on families.

### Acknowledgements

We express our sincere gratitude to **Professor Padmapriya Arasanipalai Kandhadai** for her continuous guidance and valuable feedback throughout the project development cycle. We would also like to thank the parents who participated in interviews and testing, offering insightful perspectives that directly shaped many platform features. Your input was instrumental in aligning EcoTots with real community needs.

## References

### 🔧 Technology & Development

- Firebase Documentation. https://firebase.google.com/
- React Documentation. https://reactjs.org/
- MongoDB Documentation. https://www.mongodb.com/docs/
- Node.js Official Docs. https://nodejs.org/en/docs/
- Express.js Guide. https://expressjs.com/

- Bomberbot – Building AI Chatbots with the MERN Stack. https://www.bomberbot.com/react/build-an-ai-chatbot-with-the-mern-stack-a-comprehensive-guide/
- Firebase Authentication Tutorial. https://www.freecodecamp.org/news/firebase-authentication-tutorial/
- StackOverflow – Real-Time Location with MongoDB & Node.js. https://stackoverflow.com/questions/66757297/real-time-location-system-using-mongodb-nodejs
- Medium – Google Maps API in MERN Applications. https://medium.com/@shubhamrajput252000/how-to-find-the-nearest-location-using-google-maps-in-a-mern-stack-application-81baab1cca1c

### 🤖 AI & Chatbots

- OpenAI – ChatGPT Introduction. https://openai.com/chatgpt/
- Ollama – Run and Build AI Models Locally. https://ollama.com/
- Rasa – Open Source Conversational AI. https://rasa.com/

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## Appendix A: Installation Guide

This section outlines how to set up and run the EcoTots web application on a local development environment.

### Prerequisites:

- Node.js (v18 or later)
- MongoDB Atlas account
- Firebase project with storage and authentication enabled

### Steps:

1. **Clone the Repository**
   - ➔ git clone https://github.com/lovishdhanda/W25_4495_S2_LovishD.git
   - ➔ tcd W25_4495_S2_LovishD
2. **Install Dependencies**
   - **1st terminal (Frontend)**
   - ➔ cd implementation/ecotots/client
   - ➔ npm install

- **2nd terminal (Backend)**
  - ➔ cd implementation/ecotots
  - ➔ npm install

3. **Set Up Environment Variables**

- Create .env files in both frontend and backend folders.

```
MONGO = "mongodb+srv://dhandal:dhandal@data-
ecotots.3xb2a.mongodb.net/?retryWrites=true&w=majority&appName=data-
ecoTots"
JWT_SECRET = 'QWERTYUIOP1234567890'
OPENAI_API_KEY='sk-proj-atxYjyaKF93lWBkDadRTqLMoEc3ddbHcjEQle-
_ggXfhUwT9Asil-
TdHLrqqHNg9SUoXqi0G1pT3BlbkFJJE0RhPWk0zUmhbOyqLPgUpxMiKoa0cjjf
OhSbHsgqsFsyluCgcmTP52oQsrNTRR69qK0zbl9sA'
VITE_FIREBASE_API_KEY = 'AIzaSyChNG1-u1LUAJM-Mt9PJaANN51IUaqMFAg'
```

4. **Run the Backend Server**
   - ➔ cd implementation/ecotots/client/
   - ➔ npm run dev

5. **Run the Frontend Server**
   - ➔ cd implementation/ecotots/
   - ➔ npm run dev

6. **Access the App**
   - ➔ Open your browser and go to http://localhost:5173

## Appendix B: User Guide

This guide is designed for end-users (parents/guardians) to help them navigate the EcoTots platform.

**1. Getting Started**

- Go to the homepage and create an account using Google Sign-In or email.
- Set up your profile and start browsing.

**2. Searching for Clothes**

- Use the search bar to type keywords like "t-shirt boys".

- Apply filters: Gender, Size, Condition (New/Used).

**3. Posting a Listing**

- Click “Sell Now”.
- Fill in item name, condition, size, category, and price.
- Upload up to 6 images.
- Submit your post and it will appear on the homepage.

**5. Using the Chatbot**

- Ask questions like “How do I post?” or “Show me jackets.”
- The bot will guide you or redirect you to relevant listings.

**6. Managing Your Profile**

- View or edit your listings and profile info anytime.
- You can also delete listings or update prices/images.